Milestone 3 – Local Offline Persistence Layer (Room)

72ITSE30603 Mobile Application Programming · Group 03 · Finance Tracker (Android, Kotlin, MVVM + ViewBinding)

MERGE PR	MERGE COMMIT ON MAIN	BRANCH
github.com/noway165/72ITSE30603_MobileApp_Team03/pull/8	`4d1b0593c872062b225027988e91e4d8e621ae2e`	`feature/room-database` → `main`

Offline feature: the personal finance ledger. Categories, transactions and monthly budgets are stored in the Room database `finance_tracker.db` and survive app restarts; exchange rates are cached in Room so the currency converter and reports work without a network. Pipeline: `Entity` → `DAO` (reads return `Flow`, writes are `suspend`) → `AppDatabase` thread-safe singleton (`@Volatile` + `synchronized`, opened by the `Application` class) → `Repository` → `ViewModel` (`StateFlow` UI state) → Fragment with ViewBinding. A first-run callback seeds the sample data.

Database Schema Map

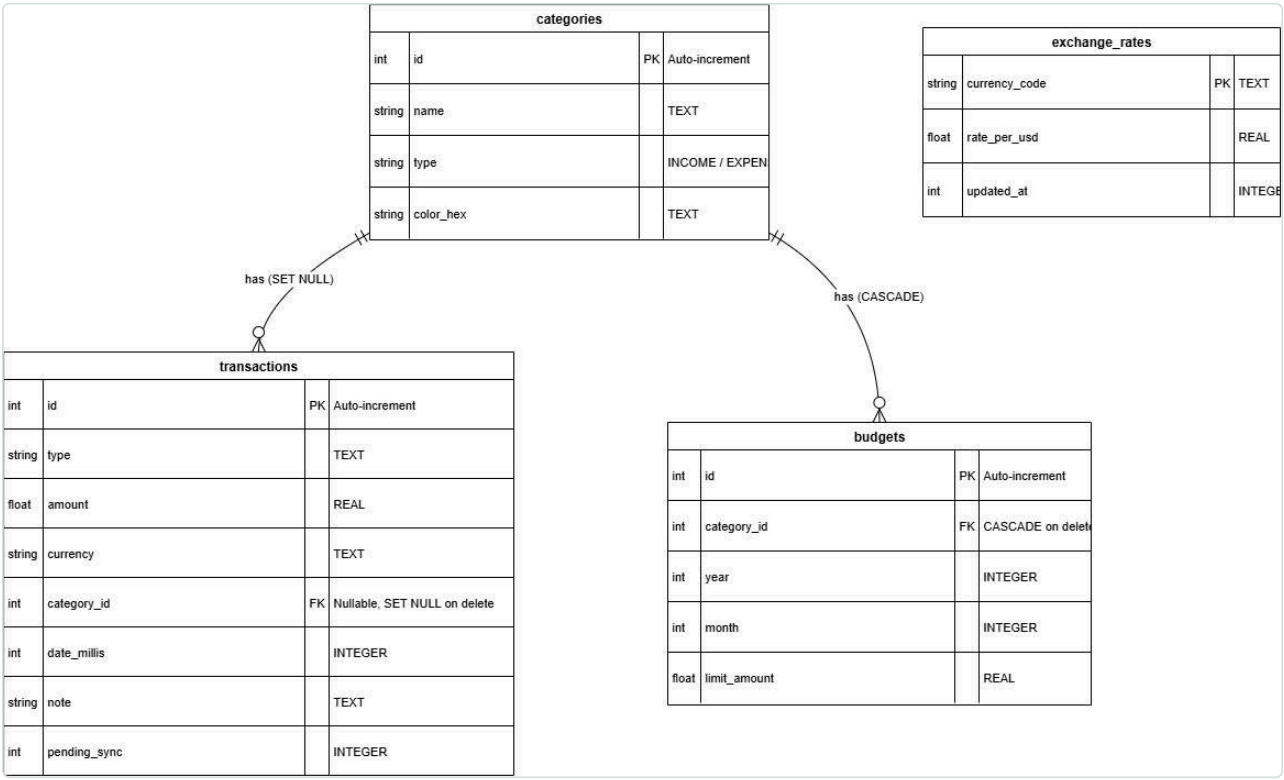
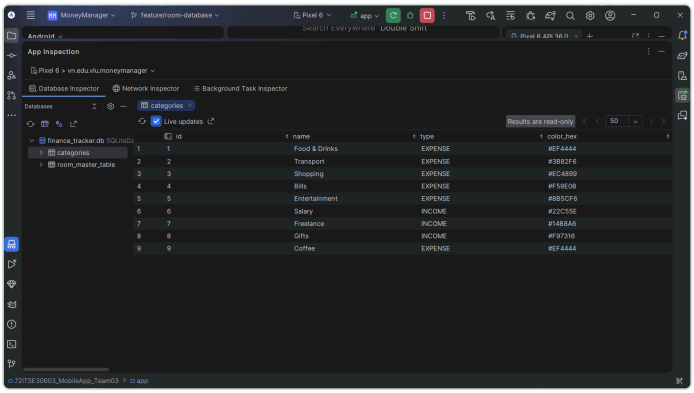
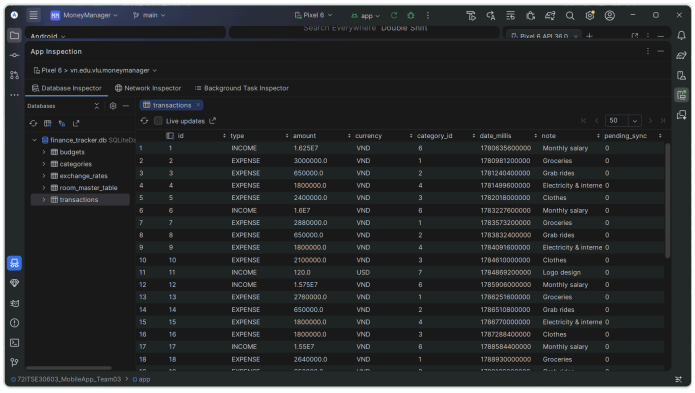


Table	Columns (SQLite type)	Keys and constraints
`categories`	id INTEGER, name TEXT, type TEXT (INCOME / EXPENSE), color_hex TEXT	PK id (auto-increment); UNIQUE (name, type)
`transactions`	id INTEGER, type TEXT, amount REAL, currency TEXT, category_id INTEGER NULL, date_millis INTEGER, note TEXT, pending_sync INTEGER	PK id; FK category_id → categories.id ON DELETE SET NULL; INDEX category_id, date_millis
`budgets`	id INTEGER, category_id INTEGER, year INTEGER, month INTEGER, limit_amount REAL	PK id; FK category_id → categories.id ON DELETE CASCADE; UNIQUE (category_id, year, month)
`exchange_rates`	currency_code TEXT, rate_per_usd REAL, updated_at INTEGER	PK currency_code

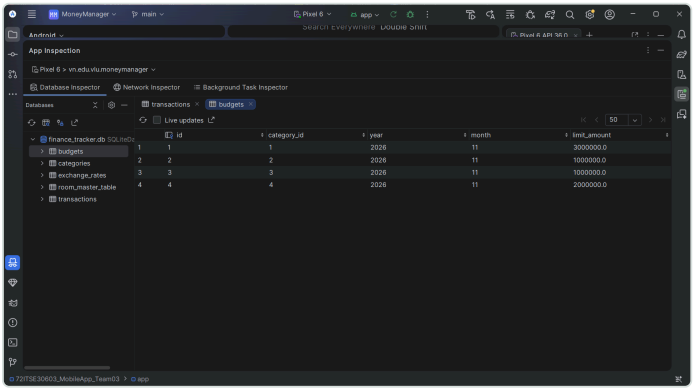
Database Inspector Proof (Android Studio, Pixel 6 API 36 emulator)



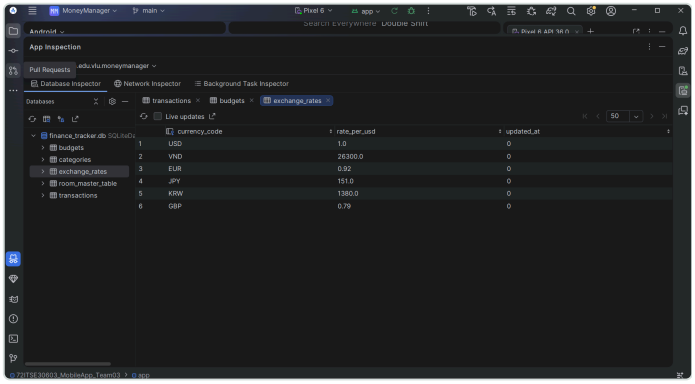
categories – 8 seeded rows + "Coffee" added in the app



transactions – 36 rows, category_id foreign key



budgets – one row per category and month



exchange_rates – offline rate cache (updated_at 0 = not downloaded yet)

Team Contribution Matrix (WBS)

Member	Tasks completed	Git evidence
Tăng Thoại Lâm (Leader)	Room + KSP setup; `AppDatabase` singleton, `FinanceTrackerApp`, type converters, seed callback; `categories` entity (UNIQUE index), `CategoryDao`, `CategoryRepository`, `CategoryViewModel` (duplicate-name check), fake DAO + tests, `CategoryDaoTest`; schema map; integration of the exchange-rate branch (conflict resolution, test fixes); final PR, Database Inspector screenshots, this report	`2d1374c`, PR #6 (`9230aff`), `c114391`, `345cead`, PR #8
Kiều Bảo Giang	`transactions` and `budgets` entities with foreign keys (SET NULL / CASCADE), indexes and UNIQUE constraint; `TransactionDao`, `BudgetDao`; repositories and ViewModels on Room; seed data for both tables; fake DAOs and unit tests; `TransactionDaoTest`, `BudgetDaoTest` (16 device tests)	PR #7 (`5783c60`)
Nguyễn Nam Khoa	`exchange_rates` entity, `ExchangeRateDao` (`@Upsert`); `CurrencyRepository` reading rates from Room and writing refreshed rates (offline cache)	`70569a7` (merged in `345cead`)

Git Workflow and Verification

`feature/room-database` was created from `main`; each member worked on a sub-branch merged into it (PR #6 categories, PR #7 transactions & budgets, exchange-rate branch merged by the leader), then PR #8 merged `feature/room-database` into `main`. Checks on the merged code: 117 JVM unit tests and 22 instrumented DAO tests pass; the app was run on the emulator and the database file was inspected (8 / 36 / 4 / 6 rows).